

Karim Hajj Ali

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EDUCATION

American University of Beirut

Expected June 2027

Bachelor of Engineering (B.E.) in Computer Science and Engineering

Beirut, Lebanon

- **Relevant Coursework:** Design & Analysis of Algorithms, Data Structures & Algorithms, Object-oriented & Effective Java Programming, Introduction to Machine Learning, AI in Industry, Database Systems, Linear Algebra, Discrete Structures, Cryptography & Network Security.

PROJECTS

CedarFix AI | *Python, Qwen2.5, LoRA, RAG, Qdrant, GCP*

Feb 2026 – May 2026

- Built a multilingual AI platform that classifies Lebanese infrastructure complaints and routes them to the correct government entity (Arabic, English, French, Arabizi).
- Gathered and curated a multilingual fine-tuning dataset covering electricity, water, roads, telecom, and waste complaint types.
- Fine-tuned Qwen2.5-3B-Instruct with LoRA (with colleagues) for structured JSON extraction, driving classification, routing, and moderation.
- Integrated RAG routing (Qdrant), CLIP image understanding, HDBSCAN clustering, and MLflow/Prometheus/Grafana on GCP/GKE.

Ou3a-Joura! | *Android sensors, ML, Python, Java*

Oct 2025 – Dec 2025

- Developed an ML-based system to detect road potholes from smartphone sensor data.
- Processed accelerometer and gyroscope time-series data collected from Android devices.
- Trained and evaluated models to classify pothole vs non-pothole events.

xv6 Operating System Extensions | *C, xv6, QEMU*

Jul 2025 – Aug 2025

- Implemented custom xv6 system calls and user-level programs to interact with kernel data structures.
- Extended the xv6 boot process by adding a login mechanism using console I/O and file descriptor management.
- Developed and integrated Unix-like user utilities (e.g., `head`) into the xv6 userland build system.

Bankruptcy Risk Prediction | *Python, CatBoost, Docker*

Sep 2025

- Built a machine learning system to predict next-year corporate bankruptcy from financial data.
- Engineered time-aware labels to prevent data leakage in year-ahead prediction.
- Trained an imbalance-aware CatBoost model with validation-based threshold optimization.

Smart Hotel Reservation System | *PostgreSQL, SQL/PLpgSQL*

Sep 2025 – Dec 2025

- Designed a relational database for a smart hotel system covering reservations, payments, services, staff, events, and multi-branch operations.
- Built an ERD and mapped it to a normalized schema with advanced relationships (M:N, weak entities, specialization, unary supervision).
- Implemented the SQL schema with constraints, indexes, views, stored procedures, and triggers to enforce business rules and data consistency.

TECHNICAL SKILLS

Programming: Python, C/C++, Java, SQL

AI/ML: scikit-learn, CatBoost, LLM fine-tuning (LoRA), RAG, Qdrant, MLflow, data preprocessing, model evaluation

Tools: GitHub, Docker, Flask, APIs

Other: Problem Solving, Team Collaboration, Communication

EXTRACURRICULAR ACTIVITIES

Member of IEEE, actively attending technical events, workshops, and networking sessions.

Member of IPark Club; won 1st place in a team-based IPark competition among 7 teams, earning a \$150 prize per team member.

ADDITIONAL INFORMATION

Languages: English (Proficient), French (Intermediate), Arabic (Native)